

Proving Chained SHA-256 Compressions with F2Z

Protocol Specification — application profile v1

Status: normative draft; profile SHA256-F2Z-MULTICOMP-APP-V1-0000; 17 August 2026

Scope. This profile proves a public chain of SHA-256 compression calls, including feed-forward. It commits only the base witness f , derives the virtual bit tensor h , and instantiates the eleven reductions ending in the F2Z PCS opening. SHA padding, private messages, and the internals of the named GKR and Ligerito subprotocols are outside this document.

1 Constraints

This section fixes the SHA application matrices. SHA-256 word semantics are those of FIPS 180-4. The row templates and the one-compression counts are those proved in [Freigen commit 93d3c55](#); this profile additionally fixes public-word binding and the canonical row sharing for a chain of N compressions.

1.1 Statement and compact matrix geometry

The public statement contains the initial and final chaining values and all message blocks:

$$x_{\text{sha}} := (N, H_0, (B_i)_{i=0}^{N-1}, H_N), \quad H_i \in (\{0, 1\}^{32})^8, \quad B_i \in (\{0, 1\}^{32})^{16}. \quad (1)$$

Every word in [Eq. \(1\)](#) has big-endian byte encoding. The canonical `app_statement_bytes` are $\text{LE}_{32}(N)$ followed by $H_0, B_0, \dots, B_{N-1}, H_N$, four big-endian bytes per word and no inner lengths. Their exact length is $68 + 64N$ bytes; trailing bytes reject. This profile requires $1 \leq N \leq 429389$, where the upper endpoint is the largest N for which $20005N + 257 \leq 2^{33}$. Before allocating, the statement preflight computes $68 + 64N$, n_f , n_h , and n_A with checked integer arithmetic and rejects on any overflow or bound failure. Padding and private-message hashing require different application profiles.

The compact Boolean input and virtual-output lengths, excluding the affine constant coordinate, are

$$n_f := 6641N + 256, \quad n_h := 20005N + 256. \quad (2)$$

Set

$$\begin{aligned} m_f &:= \max(22, \lceil \log_2 n_f \rceil), & m_h &:= \max(22, \lceil \log_2 (n_h + 1) \rceil), \\ (t_f, s, W_{\text{PCS}}) &= (m_f - 7, 7, 1), & (t_h, s, W_{\text{PCS}}) &= (m_h - 7, 7, 1). \end{aligned} \quad (3)$$

The profile requires $22 \leq m_f \leq m_h \leq 33$. Otherwise it rejects and a different profile is required. In particular the virtual chunk width is $127 - t_h - W_{\text{PCS}} = 133 - m_h \geq 100$, so $L = 1$ and $\mathcal{A} = \{0\}$. The packed-opening domain exponent is $m_p = t_f$.

Let $f \in \mathbb{F}_2^{n_f}$ be ordered as follows: all public message bits, the initial-state bits, and then the hinted decomposition bits in circuit execution order. Within a word, bit $j = 0$ is least significant and

$$[X] := \sum_{j=0}^{31} 2^j X[j], \quad \text{lo}_{32}(Q)[j] := Q[j] \quad (0 \leq j < 32). \quad (4)$$

The execution order within each compression is: rounds $0, \dots, 15$; schedule-plus-round steps $16, \dots, 61$; inlined round 62; and the fused round-63 feed-forward. Adjacent compressions reuse the previous output bits as the next state rather than allocating a duplicate state.

Let $D_f \in \mathbb{F}_2^{2^{t_f} \times 2^s}$ be the committed tensor and let J_f inject the first n_f row-major coordinates into that tensor, so its transpose selects them and $f = J_f^T \text{vec}(D_f)$. The remaining committed padding is ignored by this relation; an honest encoder zeros it. The inherited PCS packing order is the transpose of this semantic order. Define

$$\begin{aligned} B_f[c, b] &:= D_f[b, c], \\ [\Pi_f \text{vec}(D_f)]_{c2^{t_f}+b} &:= [\text{vec}(D_f)]_{b2^s+c}, & \text{vec}(B_f) &= \Pi_f \text{vec}(D_f), \end{aligned} \quad (5)$$

for $0 \leq b < 2^{t_f}$ and $0 \leq c < 2^s$. Thus the committed object is

$$O := \text{Enc}_C(\text{Pack}_{128}(B_f)), \quad \rho := \text{Root}(O). \quad (6)$$

Equation (5) is the only semantic-to-PCS index permutation. P8/V8 applies it to the adjoint coefficient, and P10/V10 uses the resulting PCS order for the ring switch.

1.2 The binary matrix M_{SHA}

Use augmented vectors $\bar{f} = (1, f)$ and $\bar{h} = (1, h_{\text{sem}})$. The Freigen matrix is the fixed binary matrix

$$M_{\text{SHA}} \in \mathbb{F}_2^{(n_h+1) \times (n_f+1)}, \quad \bar{h} = M_{\text{SHA}} \bar{f}. \quad (7)$$

Let J_h inject $\bar{h}$ into the first $n_h + 1$ coordinates of $D_h \in \mathbb{F}_2^{2^t h \times 2^s}$. Split $M_{\text{SHA}} = [m_0 \mid M_1]$ into its constant and nonconstant columns. The PCS form is

$$\text{vec}(D_h) = h_0 + T \text{vec}(D_f), \quad h_0 := J_h m_0, \quad T := J_h M_1 J_f^\top. \quad (8)$$

Only D_f is committed. Neither M_{SHA} nor T is materialized: their rows are emitted from the following parity templates.

For a word X and $0 \leq j < 32$, define

$$\begin{aligned} \Sigma_0(X)[j] &= X[j+2] \oplus X[j+13] \oplus X[j+22], \\ \Sigma_1(X)[j] &= X[j+6] \oplus X[j+11] \oplus X[j+25], \\ \sigma_0(X)[j] &= X[j+7] \oplus X[j+18] \oplus X[j+3], \\ \sigma_1(X)[j] &= X[j+17] \oplus X[j+19] \oplus X[j+10], \end{aligned} \quad (9)$$

where every rotation index is modulo 32, while the final σ_0, σ_1 shift term is zero when $j+3 \geq 32$ or $j+10 \geq 32$, respectively. The other nonlinear-function characters are

$$P_{ef} = e \oplus f, \quad P_{eg} = e \oplus g, \quad P_{abc} = a \oplus b \oplus c. \quad (10)$$

The integer combinations used below are consequently

$$C_h := [f] + [g] - [P_{ef}] + [P_{eg}] = 2 \text{Ch}(e, f, g), \quad C_m := [a] + [b] + [c] - [P_{abc}] = 2 \text{Maj}(a, b, c). \quad (11)$$

The remaining rows of M_{SHA} are identity rows for input and hinted bits. A 34-bit schedule result derives bit zero as the XOR of its four addends' low bits and commits only bits $1, \dots, 33$; a 33-bit ordinary feed-forward result does the same for its two addends. All bits of the Q^e and Q^a decompositions are committed. Concretely, for $16 \leq k \leq 61$ and $j \in \{1, 2, 3, 5, 6, 7\}$,

$$\begin{aligned} Q_k^W[0] &= W_{k-16}[0] \oplus \sigma_0(W_{k-15})[0] \oplus W_{k-7}[0] \oplus \sigma_1(W_{k-2})[0], \\ Q_j^{\text{out}}[0] &= H_i[j][0] \oplus S_{64}[j][0]. \end{aligned} \quad (12)$$

The two identities in Eq. (12) are M_{SHA} rows, not R1CS rows. In canonical row order the exact census is

row family in M_{SHA}	rows	
affine constant	1	
message inputs; initial state	$512N + 256$	
$\Sigma_0, \Sigma_1, P_{ef}, P_{eg}, P_{abc}$ in 64 rounds	$10240N$	
σ_0, σ_1 in 48 schedule steps	$3072N$	
$Q_{16}^W, \dots, Q_{61}^W$	$1564N$	
Q^e decompositions	$2242N$	
Q^a decompositions	$2177N$	
six remaining feed-forward decompositions	$198N$	
total	$20005N + 257$	(13)

The augmented column count, in the same constant-first convention, is

$$1 + (512N + 256) + 1518N + 2242N + 2177N + 192N = 6641N + 257. \quad (14)$$

For $N = 1$, Eqs. (13) and (14) give exactly 20262 rows and 6898 columns. The previous 20457×7145 system loses 64 rows/columns by coupling a^+ to e^+ , 66 by inlining W_{62}, W_{63} , 65 by fusing final feed-forward, and another 52 columns by deriving low sum bits.

1.3 The linear $\mathbb{Z}$ -R1CS rows

For compression i , let $(a_k, b_k, c_k, d_k, e_k, f_k, g_k, h_k)$ denote the state before round k , and let W_k denote its schedule word. For $0 \leq k \leq 61$, the two round rows are

$$\begin{aligned} 2[Q_k^e] &= 2([d_k] + [h_k] + [\Sigma_1(e_k)] + K_k + [W_k]) + C_h(k), \\ 2[Q_k^a] &= 2([e_{k+1}] + [\Sigma_0(a_k)] - [d_k]) + C_m(k) + 2^{33}, \end{aligned} \quad (15)$$

where Q_k^e is 35 bits, Q_k^a is 34 bits, $e_{k+1} = \text{lo}_{32}(Q_k^e)$, and $a_{k+1} = \text{lo}_{32}(Q_k^a)$. The other six words are wiring aliases, not R1CS rows: $(b_{k+1}, c_{k+1}, d_{k+1}, f_{k+1}, g_{k+1}, h_{k+1}) = (a_k, b_k, c_k, e_k, f_k, g_k)$. Equation (15) uses $a^+ = e^+ - d + \Sigma_0(a) + \text{Maj}(a, b, c) \pmod{2^{32}}$; it does not allocate a separate T_1 or 35th Q^a bit.

For $16 \leq k \leq 61$, one schedule row is also present:

$$[Q_k^W] = [W_{k-16}] + [\sigma_0(W_{k-15})] + [W_{k-7}] + [\sigma_1(W_{k-2})], \quad W_k = \text{lo}_{32}(Q_k^W), \quad (16)$$

where Q_k^W is 34 bits and its low bit is derived as specified above. Define the unmaterialized terminal sums

$$S_k := [W_{k-16}] + [\sigma_0(W_{k-15})] + [W_{k-7}] + [\sigma_1(W_{k-2})] \quad (k \in \{62, 63\}). \quad (17)$$

Round 62 has only

$$\begin{aligned} 2[Q_{62}^e] &= 2([d_{62}] + [h_{62}] + [\Sigma_1(e_{62})] + K_{62} + S_{62}) + C_h(62), \\ 2[Q_{62}^a] &= 2([e_{63}] + [\Sigma_0(a_{62})] - [d_{62}]) + C_m(62) + 2^{33}, \end{aligned} \quad (18)$$

with 36-bit Q_{62}^e and 34-bit Q_{62}^a .

Round 63 is fused with the first and fifth feed-forward words. If $A_{\text{in}} = H_i[0]$ and $E_{\text{in}} = H_i[4]$, its rows are

$$\begin{aligned} 2[Q_e^{\text{out}}] &= 2([d_{63}] + [h_{63}] + [\Sigma_1(e_{63})] + K_{63} + S_{63} + E_{\text{in}}) + C_h(63), \\ 2[Q_a^{\text{out}}] &= 2([\text{lo}_{32}(Q_e^{\text{out}})] + [\Sigma_0(a_{63})] + A_{\text{in}} - E_{\text{in}} - [d_{63}]) + C_m(63) + 2^{34}. \end{aligned} \quad (19)$$

Here Q_e^{out} is 36 bits and Q_a^{out} is 35 bits; their low 32 bits are $H_{i+1}[4]$ and $H_{i+1}[0]$. For $j \in \{1, 2, 3, 5, 6, 7\}$, the six remaining rows are

$$[Q_j^{\text{out}}] = H_i[j] + S_{64}[j], \quad H_{i+1}[j] = \text{lo}_{32}(Q_j^{\text{out}}), \quad (20)$$

where S_{64} is the final pre-feed-forward state and each Q_j^{out} is 33 bits with a derived low bit.

The internal row count per compression is therefore

$$62 \cdot 2 + 46 + 2 + 2 + 6 = 180. \quad (21)$$

This application additionally binds the sixteen public message words per compression and the eight words of each public endpoint. Intermediate states are shared. Set $n_A := 196N + 16$ and $d_A := \lceil \log_2 n_A \rceil$. With $h := \text{lift}_2(\text{vec}(D_h)) \in \mathbb{Z}^{2^{m_h}}$, the row-padded integer constraint matrix is

$$\begin{aligned} A_x &\in \mathbb{Z}^{2^{d_A} \times 2^{m_h}}, \quad A_x h = 0, \\ A_x[\beta, *] &= 0 \quad (n_A \leq \beta < 2^{d_A}). \end{aligned} \quad (22)$$

Rows are ordered blockwise as Eqs. (15), (16) and (18) to (20), followed by message pins and the two endpoint-pin families. Omitted columns of A_x are zero; an implementation stores row templates, not the padded matrix.

Finally, projection to the fixed PCS field is accepted only after the verifier recomputes

$$B_x := \max_{\beta < 2^{d_A}} \sum_z |A_x[\beta, z]|, \quad 2B_x < 2^{39} < q = 2^{100} - 15. \quad (23)$$

Equation (23) makes reduction modulo q injective on every reachable bit-witness residual. Failure is a hard reject.

2 Prover–verifier interaction

Read each table from top to bottom. A blank Prover or Verifier cell means that actor is idle. The middle column records the event order. An **N** or **H** arrow carries a NARG record or Hint; **D** marks a derived challenge or public value and carries no proof bytes.

Every NARG record is absorbed before its dependent challenge. A Hint is never absorbed and must be checked before the next squeeze. The sole exception to immediate NARG absorption is a PCS PoW nonce: parse it, permit no intervening event, verify its predicate, and then absorb it. Both proof streams and all primitive encodings are inherited from the F2Z PCS v1 profile. Tags 0x1001 and 0x1101 sample a field scalar by requesting 16 bytes with auxiliary payload $\text{LE}_{32}(a)$ for attempts $a = 0, 1, \dots$. Each raw XOF result enters the transcript before its high 28 bits are masked. The first value below q is accepted; rejection occurs before attempt 2^{32} , so the u_{32} counter never wraps.

P0/V0. Fix the input and bind the commitment

Prover P	Traffic	Verifier V
Derive $h = \text{lift}_2(h_0 + T \text{vec}(\pi_2(D_f)))$, then set $O := \text{Enc}_C(\text{Pack}_{128}(B_f))$ and Merkle-commit only D_f .	$\xrightarrow{D[\rho]}$	Validate the public SHA chain, both tensor layouts, all application, virtual-core, GKR, and code identifiers, and ρ . Require $\text{ord}(g) = 2^{128} - 1$. Absorb tag 0x0001, followed by tag 0x0002, before the first challenge.

Here T is the statement-derived SHA tap matrix and h_0 its affine column. Tag 0x0001 binds the virtual-core identifier SHA256-F2Z-VIRTUAL-CORE-V1-00000. Tag 0x0002 binds the application profile, subordinate identifiers, statement digest, shape, generator, and root. Publishing ρ does not discharge C .

OPEN CLAIMS

- S** ^{NEW}
 $A_x h = 0$ over $\mathbb{Z}$
- H** ^{NEW}
 $\pi_2(h) = h_0 + T \text{vec}(\pi_2(D_f))$
- C** ^{NEW}
 $\rho = \text{Root}(O)$

P1/V1. Multilinearize the constraint rows

Prover P	Traffic	Verifier V
On witness h , form $Q(Z) := \sum_{\beta \in \{0,1\}^{d_A}} \text{eq}(\beta, Z) \langle A_x[\beta, *], h \rangle_{\mathbb{Z}}$.		Validate the canonical row padding and bind the formal rewrite $A_x h = 0 \iff Q _{\{0,1\}^{d_A}} = 0$. No witness evaluation is performed.

P1/V1 has no message, absorb, challenge, or soundness loss.

OPEN CLAIMS

- S** ^{REDUCED}
 $Q(\beta) = 0$ on $\{0, 1\}^{d_A}$
- H**
 $\pi_2(h) = h_0 + T \text{vec}(\pi_2(D_f))$
- C**
 $\rho = \text{Root}(O)$

P2/V2. Project the residuals into $\mathbb{F}_q$

Prover P	Traffic	Verifier V
		Set $\bar{A} := \pi_q(A_x)$ and require $q > 2 \max_{\beta \in \{0,1\}^{d_A}} \sum_j A_x[\beta, j] $.

This bound makes projection injective on every reachable residual; otherwise V2 rejects.

OPEN CLAIMS

- S** ^{REDUCED}
 $\bar{A} \pi_q(h) = 0$
- H**
 $\pi_2(h) = h_0 + T \text{vec}(\pi_2(D_f))$
- C**
 $\rho = \text{Root}(O)$

P3/V3. Collapse the public constraint rows

Prover P	Traffic	Verifier V
Derive $v := \bar{A}^T \text{eq}(\cdot, r_{\text{con}})$.	$\xleftarrow{D[r_{\text{con}}]}$	Squeeze $r_{\text{con}} \in \mathbb{F}_q^{d_A}$ with tag 0x1001 in increasing coordinate order. Independently derive the same v and set the target to zero.

Each coordinate uses scope $(\perp_{32}, \perp_{32}, i)$; the reduction error is at most d_A/q .

OPEN CLAIMS

- S** ^{DISCHARGED}
 $\bar{A} \pi_q(h) = 0$
- H**
 $\pi_2(h) = h_0 + T \text{vec}(\pi_2(D_f))$
- C**
 $\rho = \text{Root}(O)$
- E₀** ^{NEW}
 $\langle v, \pi_q(h) \rangle = 0$

P4/V4. Shape the claim by sumcheck

Prover P	Traffic	Verifier V
In round i , after $r_0, \dots, r_{i-1}$, compute $g_i \in \mathbb{F}_q[X]$ with $\deg g_i \leq 2$.	$\xrightarrow{N[g_i]}$	Read and absorb tag $0x1002$ at $(\perp_{32}, \perp_{32}, i)$, then require $g_i(0) + g_i(1) = \sigma_{i-1}$. Squeeze $r_i \in \mathbb{F}_q$ under tag $0x1101$ at the same scope.
Set $\sigma_i := g_i(r_i)$.	$\xleftarrow{D[r_i]}$	Set $\sigma_i := g_i(r_i)$.
After the last round, derive $r_h = (r_0, \dots, r_{d_h-1})$, $\tau := \sigma_{d_h-1}$, $c := \widetilde{v}(r_h)$. and absorb the virtual seam under tag $0x1003$.		Derive the same (r_h, τ, c) , then absorb the virtual seam under tag $0x1003$.

Here $h_q := \pi_q(h)$, $\sigma_{-1} := 0$, and $d_h = t_h + s = m_h$; the reduction error is at most $2d_h/q$. Writing $r_h = (r_{\text{row}}, r_{\text{col}}) \in \mathbb{F}_q^{t_h} \times \mathbb{F}_q^s$, V4 derives $w_b \in \{0, \dots, q-1\}$ uniquely from $\pi_q(w_b) = \text{eq}(b, r_{\text{row}})$, and sets $w'_c = c \text{eq}(c, r_{\text{col}})$ and $y = \tau$. The derived tag $0x1003$, at scope $(\perp_{32}, \perp_{32}, \perp_{32})$, with payload

$$\text{LE}_{32}(t_h) \parallel \text{LE}_{32}(s) \parallel \text{Vec}_{\mathbb{F}_q}(r_h) \parallel \text{Enc}_{\mathbb{F}_q}(c) \parallel \text{Enc}_{\mathbb{F}_q}(\tau). \quad (24)$$

The vector uses its canonical u_{32} count and transcript order. For this virtual-core profile, tag $0x1003$ replaces **CoreSeamV1** and precedes the first tag- $0x2001$ record; every other profile rejects. G6 and G7 must close the two ledgers before this profile can accept.

P5/V5. Expose the integer column folds

Prover P	Traffic	Verifier V
For each virtual column c , compute $u_c := \sum_b w_b D_h[b, c]$, $\nu_c := g^{u_c}$.	$\xrightarrow{N[\nu]}$ $\xrightarrow{H[u]}$	Read and absorb ν under tag $0x2001$. Read u under tag $0x2081$; before any squeeze, require $0 \leq u_c < 2^{127}$ and $g^{u_c} = \nu_c$ for every c , together with $\sum_c w'_c \pi_q(u_c) = y$. Then clear the Hint.

Both vectors have $2^s = 128$ entries and scope $(0, \perp_{32}, \perp_{32})$; V0 already checked the order of g .

P6/V6. Move the folds into products

Prover P	Traffic	Verifier V
Form the witness leaf values $L[b, c] := 1 + (g^{w_b} - 1)D_h[b, c]$.	$\xleftarrow{D[\zeta]}$	After V5 clears its Hint, instantiate the corresponding product relation. Squeeze $\zeta \in \mathbb{F}_{2^{128}}^s$ under tag $0x2101$ in increasing coordinate order.
Derive $e_0 := \widetilde{\nu}(\zeta)$ and initialize the product forest.		Independently derive $e_0 := \widetilde{\nu}(\zeta)$.

V6 obtains $\zeta \in \mathbb{F}_{2^{128}}^s$ with the inherited $\mathbb{F}_{2^{128}}$ sampler and tag $0x2101$, scope $(0, \perp_{32}, i)$, in increasing $0 \leq i < s$. The exponent range makes the product reduction exact; random column compression is charged to the partial-core error.

OPEN CLAIMS

- | **H**
 $\pi_2(h) = h_0 + T \text{vec}(\pi_2(D_f))$
- | **C**
 $\rho = \text{Root}(O)$
- | **E_0 DISCHARGED**
 $\langle v, \pi_q(h) \rangle = \emptyset$
- | **E_{eq} NEW**
 $\widetilde{ch}_q(r_h) = \tau$

OPEN CLAIMS

- | **H**
 $\pi_2(h) = h_0 + T \text{vec}(\pi_2(D_f))$
- | **C**
 $\rho = \text{Root}(O)$
- | **E_{eq} DISCHARGED**
 $\widetilde{ch}_q(r_h) = \tau$
- | **T NEW**
 $u_c = \langle w, D_h[*, c] \rangle_{\mathbb{Z}}$

OPEN CLAIMS

- | **H**
 $\pi_2(h) = h_0 + T \text{vec}(\pi_2(D_f))$
- | **C**
 $\rho = \text{Root}(O)$
- | **T DISCHARGED**
 $u_c = \langle w, D_h[*, c] \rangle_{\mathbb{Z}}$
- | **P NEW**
 $\prod_b L[b, c] = \nu_c$

P7/V7. Reduce the merged product forest to LIN

Prover P	Traffic	Verifier V
Run the registered virtual-GKR prover on the product forest.	$\xrightarrow{N[\pi_{\text{GKR}}]}$	Replay the registered virtual-GKR ledger, checking and absorbing every message before its dependent challenge. Require complete consumption and return (R, e) .

The registered profile owns NARG tags $0x3001-0x30ff$ and squeeze tags $0x3100-0x31ff$; tag $0x3081$ is forbidden. It returns

$$E_H : \langle R, h_{\mathbb{F}_{2^{128}}} \rangle_{\mathbb{F}_{2^{128}}} = e, \quad h_{\mathbb{F}_{2^{128}}} = h_{0, \mathbb{F}_{2^{128}}} + T_{\mathbb{F}_{2^{128}}} f_{\mathbb{F}_{2^{128}}}, \quad (25)$$

where R is verifier-derived rather than transmitted densely. **G6** requires the exact P5–P7 ledger, reduction proof, and error; an unknown ledger rejects.

P8/V8. Apply the full virtual adjoint

Prover P	Traffic	Verifier V
Derive $a := \Pi_f T_{\mathbb{F}_{2^{128}}}^* R$, $\mu_f := e - \langle R, h_{0, \mathbb{F}_{2^{128}}} \rangle_{\mathbb{F}_{2^{128}}}$.		Independently derive the same (a, μ_f) and the public evaluator $\tilde{a}(r) = \langle R, T_{\mathbb{F}_{2^{128}}} (\Pi_f^{-1} \text{eq}(\cdot, r)) \rangle_{\mathbb{F}_{2^{128}}}$.

For the committed bits $F := \Pi_f f_{\mathbb{F}_{2^{128}}}$ in PCS (c, b) order,

$$a := \Pi_f T_{\mathbb{F}_{2^{128}}}^* R, \quad \mu_f := e - \langle R, h_{0, \mathbb{F}_{2^{128}}} \rangle_{\mathbb{F}_{2^{128}}}, \quad E_f : \langle a, F \rangle_{\mathbb{F}_{2^{128}}} = \mu_f. \quad (26)$$

The table gives its exact public evaluator. A sparse evaluator is conforming only if it is identical to the full adjoint; a columnwise shortcut rejects.

P9/V9. Run exactly one LinCheck

Prover P	Traffic	Verifier V
In round i , after $r_0, \dots, r_{i-1}$, compute $g_i \in \mathbb{F}_{2^{128}}[X]$ with $\deg g_i \leq 2$.	$\xrightarrow{N[g_i]}$ $\xleftarrow{D[r_i]}$	Read and absorb tag $0x4002$ at $(\perp_{32}, \perp_{32}, i)$, then require $g_i(0) + g_i(1) = \sigma_{i-1}$. Squeeze $r_i \in \mathbb{F}_{2^{128}}$ under tag $0x4103$ at the same scope. Set $\sigma_i := g_i(r_i)$.

Set $\sigma_i := g_i(r_i)$.

Start with $\sigma_{-1} := \mu_f$. At the final point $r = (r_{\text{lo}}, r_{\text{hi}}) \in \mathbb{F}_{2^{128}}^7 \times \mathbb{F}_{2^{128}}^{t_f}$,

$$\sigma_{m_f-1} = \tilde{a}(r) \tilde{F}(r). \quad (27)$$

P10/V10 checks Equation (27) with the first ring vector. This is the only LinCheck, uses no division, and has error at most $2m_f/|\mathbb{F}_{2^{128}}|$.

OPEN CLAIMS

- H**
 $\pi_2(h) = h_0 + T \text{vec}(\pi_2(D_f))$
- C**
 $\rho = \text{Root}(O)$
- P** **DISCHARGED**
 $\prod_b L[b, c] = \nu_c$
- E_H** **NEW**
 $\langle R, h_{\mathbb{F}_{2^{128}}} \rangle_{\mathbb{F}_{2^{128}}} = e$

OPEN CLAIMS

- H** **DISCHARGED**
 $\pi_2(h) = h_0 + T \text{vec}(\pi_2(D_f))$
- C**
 $\rho = \text{Root}(O)$
- E_H** **DISCHARGED**
 $\langle R, h_{\mathbb{F}_{2^{128}}} \rangle_{\mathbb{F}_{2^{128}}} = e$
- E_f** **NEW**
 $\langle a, F \rangle_{\mathbb{F}_{2^{128}}} = \mu_f$

OPEN CLAIMS

- C**
 $\rho = \text{Root}(O)$
- E_f** **DISCHARGED**
 $\langle a, F \rangle_{\mathbb{F}_{2^{128}}} = \mu_f$
- LC** **NEW**
 $\sigma = \tilde{a}(r) \tilde{F}(r)$

P10/V10. Discharge LinCheck and ring-switch

Prover P	Traffic	Verifier V
For $0 \leq v < 128$, compute $s_v := (B_f)_{v, \mathbb{F}_{2^{128}}}(r_{hi})$.	$\xrightarrow{N[s]}$	Read and absorb s under tag $0x4001$ at $(0, \perp_{32}, \perp_{32})$. Derive $z := \sum_{v=0}^{127} \text{eq}(v, r_{lo}) s_v = \tilde{F}(r)$ and require $\sigma_{m_f-1} = \tilde{a}(r)z$. Only after that check, squeeze $r'' \in \mathbb{F}_{2^{128}}^7$ under tag $0x4101$.
Derive the ring-switch successor (B_0, β_0) .	$\xleftarrow{D[r'']}$	Independently derive the same (B_0, β_0) . Squeeze $\eta_0 \in \mathbb{F}_{2^{128}}$ under tag $0x4102$.
Set $B := \eta_0 B_0$ and $\beta := \eta_0 \beta_0$.	$\xleftarrow{D[\eta_0]}$	Independently derive the same (B, β) .

OPEN CLAIMS

C
 $\rho = \text{Root}(O)$
LC DISCHARGED
 $\sigma = \tilde{a}(r) \tilde{F}(r)$
EP NEW
 $\langle B, P \rangle_{\mathbb{F}_{2^{128}}} = \beta$

$$B := \eta_0 B_0, \quad \beta := \eta_0 \beta_0, \quad E_P : \langle B, P \rangle_{\mathbb{F}_{2^{128}}} = \beta, \quad P = \text{Pack}_{128}(B_f). \quad (28)$$

No extra adapter message, target, or second LinCheck exists. The event rules are now fixed by the PCS profile; **G7** is the remaining implementation and full-fixture closure for this virtual profile.

P11/V11. Open the packed claim

Prover P	Traffic	Verifier V
Require $m_p = t_f$, $B : \{0, 1\}^{t_f} \rightarrow \mathbb{F}_{2^{128}}$, derive $\text{payload}_{5001} := \text{LE}_{32}(t_f) \parallel \text{Enc}_{\mathbb{F}_{2^{128}}}(\beta)$, and absorb tag $0x5001$.		Independently require the same domain and absorb the same derived tag.
Run the inherited recursive opening prover for (ρ, B, β) .	$\xrightarrow{N/H[\pi_{\text{open}}]}$	Replay the inherited opening ledger against (ρ, B, β) . Accept only if the opening succeeds, every Hint is authenticated, and both streams, all nested slices, and the host container end exactly.

OPEN CLAIMS

C DISCHARGED
 $\rho = \text{Root}(O)$
EP DISCHARGED
 $\langle B, P \rangle_{\mathbb{F}_{2^{128}}} = \beta$

The composed soundness error is

$$\varepsilon_{\text{SHA-F2Z}} \leq \frac{d_A + 2d_h}{q} + \varepsilon_{\text{partial}} + \frac{2m_f}{2^{128}} + \varepsilon_{\text{ring}} + \varepsilon_{\text{batch}} + \varepsilon_{\text{open}}. \quad (29)$$

Here $\varepsilon_{\text{partial}}$ includes P6 column compression and P7 GKR; $2m_f/2^{128}$ is the single P9 LinCheck term. Equation (29) is symbolic until **G6** and **G7** publish their proofs and fixtures. P1, P2, P5, and P8 are exact under their stated reject gates.

3 Verifier, status, and gaps

3.1 Verifier algorithm

Algorithm 1 replays P0–P11. The virtual partial core returns Eq. (25); the public adjoint, one LinCheck, and inherited ring switch then follow Eqs. (26) to (28). The direct PCS profile does not enable this virtual path and rejects on Line 4.

Algorithm 1 SHA front-end verifier and virtual-PCS handoff

Require: $\text{pp}_{\text{SHA}}, x_{\text{sha}}, \rho, \pi_{\text{host}}$
Ensure: true or false
1: $(p_{\text{core}}, p_{\text{app}}, \text{PGKR}, p_{\text{code}}, g) \leftarrow \text{PROFILEFIELDS}(\text{pp}_{\text{SHA}})$
2: **require** $p_{\text{core}} = \text{SHA256-F2Z-VIRTUAL-CORE-V1-00000} \wedge p_{\text{app}} = \text{SHA256-F2Z-MULTICOMP-APP-V1-0000} \wedge \text{ord}(g) = 2^{128} - 1 \wedge \text{PREFLIGHTSHASTATEMENT}(x_{\text{sha}}) = 1$
3: $p_{\text{part}} \leftarrow \text{ResolvePartialProfile}(p_{\text{core}}, \text{PGKR}); p_{\text{LC}} \leftarrow \text{ResolveLinCheckProfile}(p_{\text{core}}); p_{\text{ring}} \leftarrow \text{ResolveRingProfile}(p_{\text{core}}); p_{\text{open}} \leftarrow \text{ResolveCodeOpenProfile}(p_{\text{core}}, p_{\text{code}})$
4: **require** $p_{\text{part}} \neq \perp \wedge p_{\text{LC}} \neq \perp \wedge p_{\text{ring}} \neq \perp \wedge p_{\text{open}} \neq \perp$
5: derive $n_f \leftarrow 6641N + 256, n_h \leftarrow 20005N + 256, m_f \leftarrow \max(22, \lceil \log_2 n_f \rceil), m_h \leftarrow \max(22, \lceil \log_2(n_h + 1) \rceil)$
6: $t_f \leftarrow m_f - 7, t_h \leftarrow m_h - 7, n_A \leftarrow 196N + 16, d_A \leftarrow \lceil \log_2 n_A \rceil, d_h \leftarrow m_h$
7: **require** $22 \leq m_f \leq m_h \leq 33; b_{\text{host}} \leftarrow \text{PROFILEBOUNDS}(p_{\text{part}}, p_{\text{LC}}, p_{\text{ring}}, p_{\text{open}}, N, m_f, m_h);$ **require** $b_{\text{host}} \neq \perp$
8: $(c_N, c_H) \leftarrow \text{DECODEHOSTV1}(\pi_{\text{host}}, b_{\text{host}}, p_{\text{core}})$
9: derive canonical $(M_{\text{SHA}}, T, h_0, A_x, \Pi_f); B_x \leftarrow \max_{\beta < 2^{d_A}} \sum_j |A_x[\beta, j]|$
10: **require** $\text{SHAPEANDCENSUS}(M_{\text{SHA}}, T, h_0, A_x, \Pi_f, n_f, n_h, n_A) = 1$
11: **require** $2B_x < 2^{39} < q$
12: $d_{\text{app}} \leftarrow \text{APPDIGESTV1}(p_{\text{app}}, x_{\text{sha}})$
13: $\text{tr} \leftarrow \text{ABSORBPROFILEANDSTATEMENTV1}(p_{\text{core}}, p_{\text{app}}, \text{PGKR}, p_{\text{code}}, d_{\text{app}}, (t_f, 7, 1), g, \rho)$
14: $r_{\text{con}} \leftarrow \text{SQUEEZECONSTRAINTPOINT}(\text{tr}, d_A); v \leftarrow \pi_q(A_x)^{\top} \text{eq}(\cdot, r_{\text{con}})$
15: $(\text{ok}, \text{tr}, r_h, \tau) \leftarrow \text{VERIFYEVALSHAPESTREAM}(v, \text{tr}, c_N, d_h)$
16: **require** $\text{ok} = 1 \wedge |r_h| = d_h; c \leftarrow v(r_h), a_h[z] \leftarrow c \text{eq}(z, r_h)$
17: $\mathcal{I}_H \leftarrow (T, h_0, \Pi_f, t_h, 7, r_h, c, a_h, \tau, \rho, m_f, m_h, p_{\text{core}}, \text{PGKR}, g)$
18: **require** $\text{VIRTUALPROFILESADMITINSTANCE}(p_{\text{part}}, p_{\text{LC}}, p_{\text{ring}}, \mathcal{I}_H) = 1$
19: $\text{tr} \leftarrow \text{ABSORBVIRTUALSEAM}(\text{tr}, 0\text{x}1003, t_h, 7, r_h, c, \tau)$
20: $(\text{ok}, \text{tr}, R, e) \leftarrow \text{VERIFYVIRTUALPARTIALSTREAM}(p_{\text{part}}, \mathcal{I}_H, \text{tr}, c_N, c_H)$
21: **require** $\text{ok} = 1; \mu_f \leftarrow e - \langle R, h_{0, \mathbb{F}_{2^{128}}} \rangle; \text{Eval}_a(r) \leftarrow \langle R, T_{\mathbb{F}_{2^{128}}}(\Pi_f^{-1} \text{eq}(\cdot, r)) \rangle$
22: $(\text{ok}, \text{tr}, B, \beta) \leftarrow \text{VERIFYLINCHECKRINGSTREAM}(p_{\text{LC}}, p_{\text{ring}}, \text{Eval}_a, \mu_f, \rho, \text{tr}, c_N, c_H)$
23: **require** $\text{ok} = 1 \wedge \text{dom}(B) = \{0, 1\}^{t_f}$
24: $\text{tr} \leftarrow \text{ABSORBOPENINGTARGET}(\text{tr}, 0\text{x}5001, \text{LE}_{32}(t_f) \parallel \text{Enc}_{\mathbb{F}_{2^{128}}}(\beta))$
25: $\text{ok} \leftarrow \text{VERIFYPACKEDOPENSTREAM}(p_{\text{open}}, \rho, B, \beta, \text{tr}, c_N, c_H)$
26: **require** $\text{NOPENDINGHINT} = 1 \wedge \text{EXACTNARGHINTCOUNTSANDBYTES}(c_N, c_H) = 1 \wedge \text{NESTEDCURSORANDHOSTEOF}(\pi_{\text{host}}, c_N, c_H) = 1$
27: **return** ok

3.2 Provenance and current status

Quantity	Provenance and verifier treatment	Status
Public/derived	Validate x_{sha}, ρ , all IDs, g , shape, and d_{app} ; recompute maps and challenges from the bound prefix.	math specified; fixture open
Application stream	NARG round polynomials are checked and absorbed before their dependent challenge.	specified
Core/opening	The partial core is unresolved. The adjoint and one-LinCheck ring contract are specified; their integrated implementation and fixtures are still absent.	open

3.3 Gap register: do not silently resolve in code

	Source omission and this profile's rule	Owner; reviewer	Required closure evidence
G1	FIPS specifies hashing and padding, while this profile proves public chained compressions only. Padding, private blocks, and an empty chain reject.	John Wu; Shreyas Londhe	Public-statement fixtures for one and multiple blocks plus negative scope tests.
G2	The sources do not prove that fixed q is injective on every reachable integer residual. The verifier enforces $q > 2B_x$.	John Wu; Albert	Reviewed symbolic residual bound and executable maximum-row check.
G3	Earlier SHA material omitted feed-forward from its census. This profile includes every feed-forward equation and uses $n_A = 196N + 16$.	John Wu; Albert	FIPS known-answer vectors and an independently recomputed census for several chain lengths.
G4	Prototype paths expose incompatible layouts. This profile has one execution-ordered, LSB-first compact tensor. Nonsemantic filler is committed but unconstrained; honest encoders zero it.	John Wu; Shreyas Londhe	Index-map fixture, honest-zero encoding bytes, and independent root vector.
G5	Freigen proves one compression, but chained row sharing, the padded selector, h -only normalization, and the dual adjoint evaluator remain derived.	John Wu; Albert	Chain equivalence proof and constraint-by-constraint differential tests, including $\tilde{a}(r) = \langle R, T_{\mathbb{F}_{2^{128}}}(\Pi_f^{-1} \text{eq}(\cdot, r)) \rangle$.
G6	P5–P7 have no registered virtual partial-core/GKR ledger or numeric reduction error. The direct PCS profile therefore rejects.	John Wu; Shreyas Londhe	Profile ID, typed reduction proof, byte ledger, numeric error, and positive/negative fixtures.
G7	The PCS now fixes one LinCheck and its reuse of the inherited ring vector, but no integrated implementation or independent full fixture exists. Interoperability is unclaimed.	John Wu; Shreyas Londhe	Two independent verifiers that agree on full proofs and reject round, tag, order, terminal, suffix, root, and opening tampering.